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## CS771: Minor Assignment 3

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### Task 1: Knee Point Analysis

Keeping  $n_{components} = 4$  as in the lecture code, we varied the number of EM iterations  $n_{iter}$  and plotted the test MAE against  $n_{iter}$ .

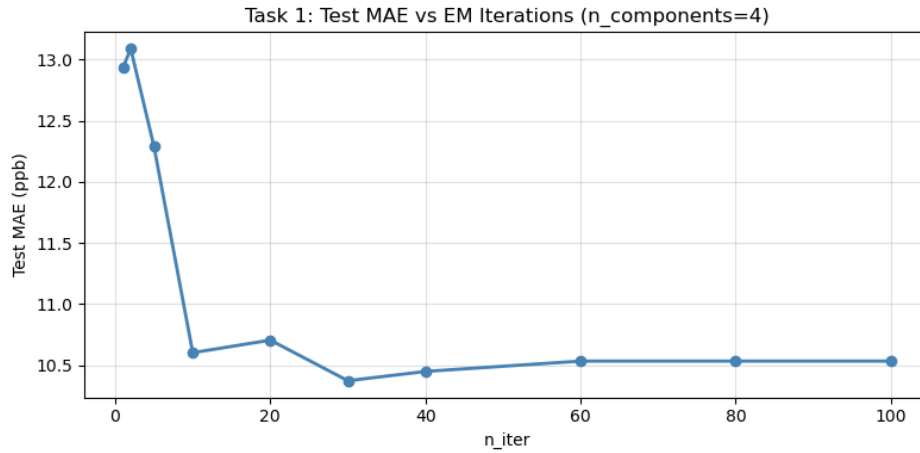


Figure 1: Test MAE vs EM iterations ( $n_{components} = 4$ )

We experimented with:

$$n_{iter} = [1, 2, 5, 10, 20, 30, 40, 60, 80, 100]$$

From the plot, we observe that:

- There is a sharp decrease in test MAE in the range  $n_{iter} = 1$  to 10, indicating rapid learning in the early EM iterations.
- In the range  $n_{iter} = 10$  to 20, the MAE shows a slight increase before stabilizing, reflecting minor instability due to the EM objective optimizing likelihood rather than MAE.
- Beyond  $n_{iter} \approx 20$ , the test MAE stabilizes with only minor variations (less than 1% change between successive values), indicating convergence.

Although the absolute minimum MAE of 10.3722 ppb is observed at  $n_{iter} = 30$ , the key observation is that the improvement beyond  $n_{iter} = 20$  is negligible (less than 1%).

Hence, the knee point is:

$$n_{iter} \approx 20$$

This suggests that using more than 20 iterations increases computational cost without significant performance gains.

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## Task 2: Effect of $n\_components$ on Test MAE

We fixed  $n\_iter = 20$  (knee point from Task 1) and varied:

$$n\_components = \{1, 2, 4, 8, 16, 32\}$$

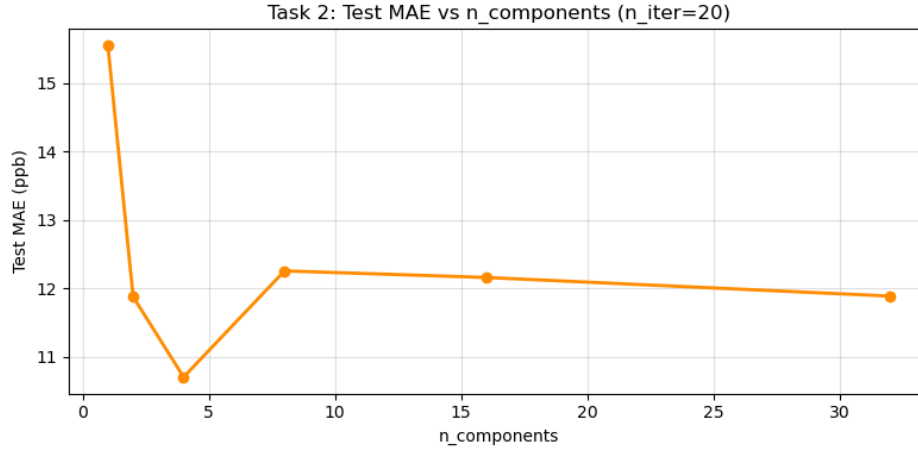


Figure 2: Test MAE vs number of components

The observed MAE values are:

$$\begin{aligned} 1 &\rightarrow 15.5517 \\ 2 &\rightarrow 11.8781 \\ 4 &\rightarrow 10.7045 \\ 8 &\rightarrow 12.2572 \\ 16 &\rightarrow 12.1610 \\ 32 &\rightarrow 11.8887 \end{aligned}$$

We observe the following trends:

- For  $n\_components = 1$ , the model reduces to a single global regression model with no auxiliary classifier, yielding the highest MAE of 15.55 ppb.
- Increasing the number of components from 1 to 4 significantly reduces MAE, indicating that multiple components help capture heterogeneity in the data.
- At  $n\_components = 8$ , performance degrades slightly, suggesting instability or suboptimal partitioning of data.
- Performance at  $n\_components = 16$  and  $32$  is comparable to  $n\_components = 2$ , suggesting diminishing returns beyond 4 components.
- The best performance is achieved at  $n\_components = 4$  with a MAE of 10.7045 ppb.

Thus, we observe a trade-off:

- Too few components  $\rightarrow$  underfitting
- Too many components  $\rightarrow$  overfitting

In this case,  $n\_components = 4$  provides the best performance.

## Task 3: Interpretation of Components in Mixture Model

To understand how the mixture model partitions the data, we analyze how different components correspond to different ranges of the target variable Ref. O3(ppb). We focus on cases where  $n\_components < 10$ , specifically  $n\_components = 2, 4, 8$ .

## Observations

### Case 1: $n\_components = 2$

With only two components, the model performs a coarse partitioning of the data. One component primarily captures very low O<sub>3</sub> values (approximately 5–20 ppb), while the other component captures a wide range of medium to high values (approximately 20–120+ ppb). This indicates underfitting, as the model is unable to distinguish finer variations in pollution levels.

### Case 2: $n\_components = 4$

With four components, the model produces a much more refined partitioning of the data:

- Very low O<sub>3</sub> values (approximately 5–15 ppb)
- Low-medium O<sub>3</sub> values (approximately 15–40 ppb)
- Medium O<sub>3</sub> values (approximately 40–70 ppb)
- High O<sub>3</sub> values (approximately 70–120+ ppb)

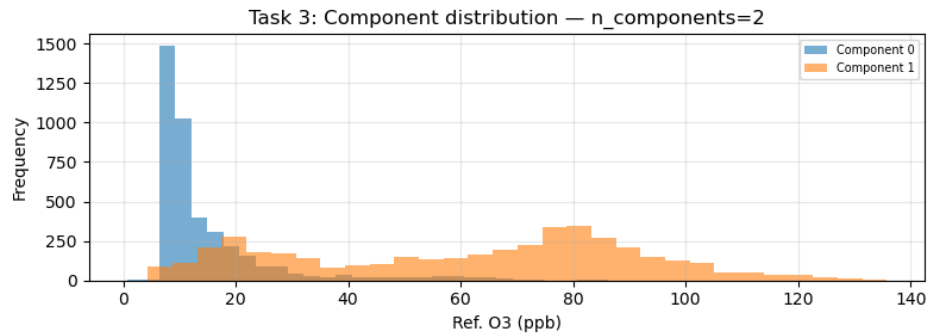
This clearly shows that the model separates the data into meaningful pollution regimes. The components are well-separated and interpretable, similar to the behavior described in the lecture.

### Case 3: $n\_components = 8$

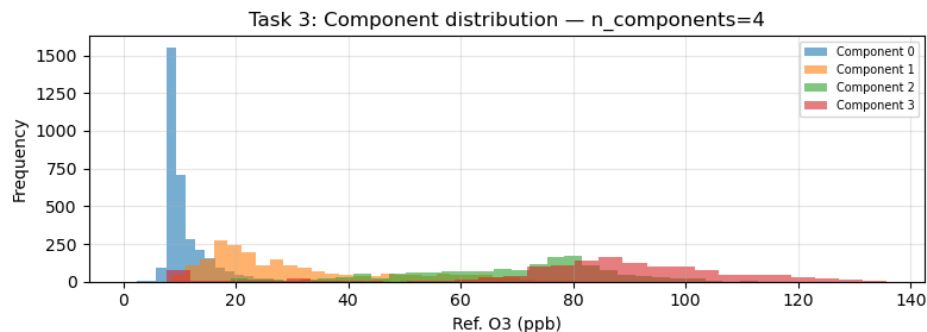
With eight components, the model creates even finer partitions. However, instead of cleanly separating new regimes, multiple components begin to overlap and represent similar ranges of O<sub>3</sub> values. Several components capture slightly different variations within the same region.

This indicates over-partitioning, where the model increases complexity without gaining clear interpretability.

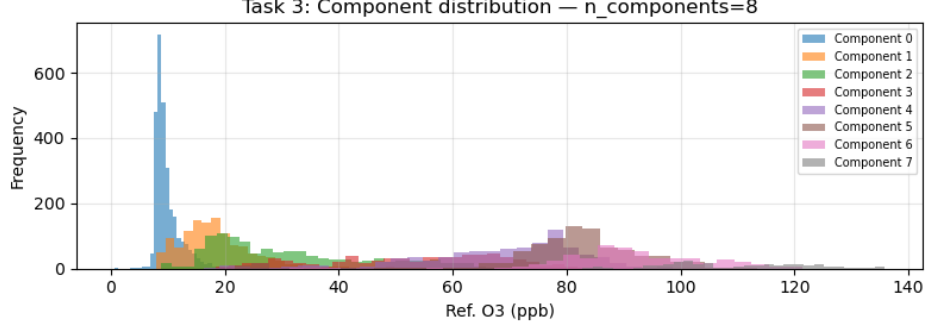
## Visualization



(a)  $n\_components = 2$



(b)  $n\_components = 4$



(c)  $n\_components = 8$

Figure 3: Component-wise distribution of Ref. O3(ppb) for  $n\_components = 2, 4, 8$

### Discussion

From these observations, we conclude the following:

- With very few components (e.g.,  $n\_components = 2$ ), the model underfits and produces overly coarse partitions of the data.
- With a moderate number of components (e.g.,  $n\_components = 4$ ), the model learns meaningful and interpretable partitions corresponding to different pollution levels.
- With a larger number of components (e.g.,  $n\_components = 8$ ), the model begins to over-partition the data, leading to overlapping components and reduced interpretability.

Thus, there exists a trade-off between model simplicity and expressiveness. A moderate number of components provides the best balance between interpretability and predictive power.

### Task 4: Feature Importance (Ablation Study)

To determine which features contribute the most and least to the model, we systematically omitted one feature at a time and observed the resulting Test MAE. We fixed  $n\_iter = 20$  and varied:

$$n\_components = \{1, 2, 4, 8, 16, 32\}$$

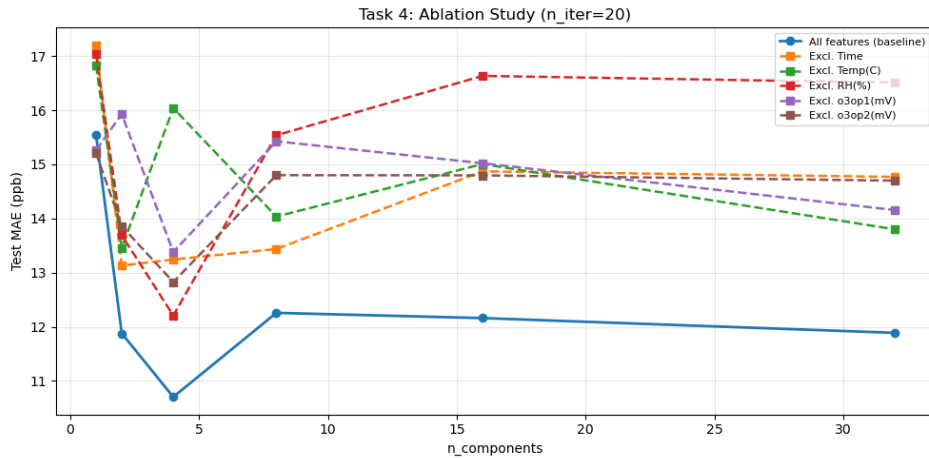


Figure 4: Test MAE vs number of components with omitted features

Comparing the best observed MAE values for each omitted feature against our baseline:

Baseline (All Features)  $\rightarrow 10.7045$   
Skipping Temp(C)  $\rightarrow 13.4567$  ( $\Delta = +2.7522$ )  
Skipping o3op1(mV)  $\rightarrow 13.3807$  ( $\Delta = +2.6762$ )  
Skipping Time  $\rightarrow 13.1310$  ( $\Delta = +2.4266$ )  
Skipping o3op2(mV)  $\rightarrow 12.8269$  ( $\Delta = +2.1224$ )  
Skipping RH(%)  $\rightarrow 12.2095$  ( $\Delta = +1.5051$ )

We rank features by their  $\Delta$  MAE (the increase in best MAE when the feature is excluded): a larger  $\Delta$  implies the model relied more heavily on that feature.

We observe the following trends:

- When **Temp(C)** or **o3op1(mV)** is removed, the Test MAE increases the most ( $\Delta \approx +2.75$  and  $+2.68$  respectively), indicating these are the most important features for prediction.
- **Time** and **o3op2(mV)** also contribute meaningfully, with  $\Delta \approx +2.43$  and  $+2.12$  respectively.
- Conversely, when **RH(%)** is removed, the Test MAE increases the least ( $\Delta = +1.51$ ), indicating it contributes least to the model's predictive accuracy among the five features.

Thus, based on the ablation study, we conclude:

- **Most Contribution**  $\rightarrow$  Temp(C)
- **Least Contribution**  $\rightarrow$  RH(%)

## Task 5: Comparison with Decision Tree Regressor

Melbae pointed out that the mixture of regressions model can be interpreted as a two-layer regression tree, where a classifier routes each data point to one of several regression models. To evaluate whether a direct regression tree can perform better, we trained a DecisionTreeRegressor and compared it with the EM-based mixture regression model.

### Experimental Setup

We used all five features: Time, Temp(C), RH(%), o3op1(mV), and o3op2(mV), with the same train-test split as used in earlier parts (first 80% timestamps for training and the rest for testing).

The Decision Tree model was trained using the following hyperparameters (tuned via grid search over  $\text{criterion} \in \{\text{squared\_error}, \text{friedman\_mse}, \text{absolute\_error}\}$  and  $\text{splitter} \in \{\text{best}, \text{random}\}$ ):

- $\text{criterion} = \text{squared\_error}$
- $\text{splitter} = \text{random}$
- $\text{max\_depth} \in \{2, 3, 4, 5, 6, 8, 10\}$

For each value of  $\text{max\_depth}$ , we recorded:

- Test MAE
- Total prediction time over the entire test set

For the mixture model, we reused results from Part 2 with:

$$n\_components = \{1, 2, 4, 8, 16, 32\}, \quad n\_iter = 20$$

Prediction time was measured using the wall-clock time required to generate predictions for the entire test dataset.

### Results

#### Decision Tree Results:

$\text{max\_depth} = 2 \rightarrow \text{MAE} = 24.3292$   
 $\text{max\_depth} = 3 \rightarrow \text{MAE} = 17.5001$   
 $\text{max\_depth} = 4 \rightarrow \text{MAE} = 17.0431$   
 $\text{max\_depth} = 5 \rightarrow \text{MAE} = 15.3726$   
 $\text{max\_depth} = 6 \rightarrow \text{MAE} = 11.0775$   
 $\text{max\_depth} = 8 \rightarrow \text{MAE} = 11.9358$   
 $\text{max\_depth} = 10 \rightarrow \text{MAE} = 11.0118$

Prediction times for Decision Trees were in the range of 0.53 to 0.90 milliseconds.

#### Mixture Model Results:

1  $\rightarrow$  MAE = 15.5517

2  $\rightarrow$  MAE = 11.8781

4  $\rightarrow$  MAE = 10.7045

8  $\rightarrow$  MAE = 12.2572

16  $\rightarrow$  MAE = 12.1610

32  $\rightarrow$  MAE = 11.8887

Prediction times for the vectorised mixture models were in the range of 0.18 to 7.22 milliseconds, with the single-component model being the fastest (0.18 ms) and the 32-component model the slowest (7.22 ms).

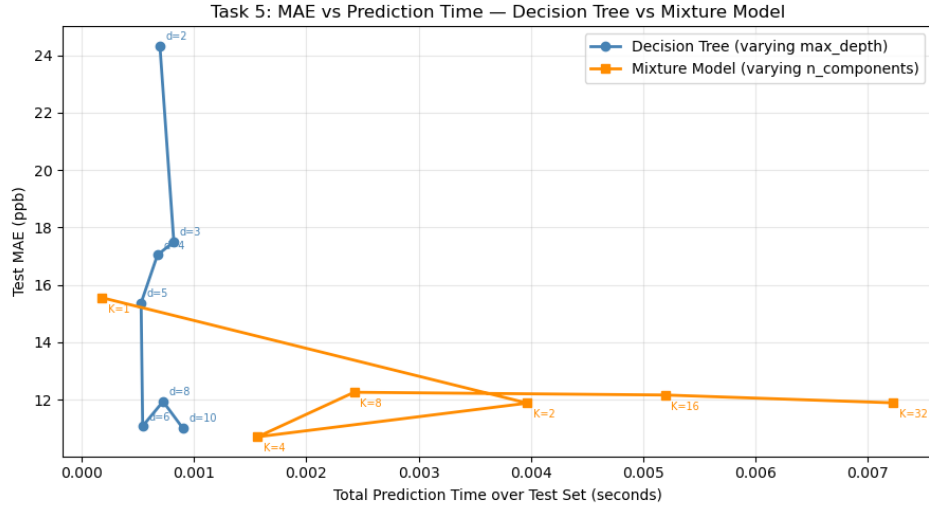


Figure 5: Test MAE vs Prediction Time for Decision Tree and Mixture Model

#### Discussion

From the comparison, we observe the following:

- **Speed:** Both models are now comparable in speed thanks to the vectorised mixture model inference. Decision Trees range from 0.53 to 0.90 ms, while mixture models range from 0.18 to 7.22 ms. The single-component mixture model is actually the fastest overall, and small- $K$  mixture models are as fast as or faster than decision trees.
- **Accuracy:** The mixture model achieves the lowest MAE of 10.70 ppb at  $n\_components = 4$ , outperforming the best Decision Tree (11.01 ppb at  $max\_depth = 10$ ). Shallow decision trees ( $max\_depth \leq 5$ ) perform significantly worse, with MAE as high as 24.33 ppb.
- **Model Behavior:** Decision Trees perform hard partitioning of the feature space, which can limit flexibility. In contrast, mixture models learn multiple specialized regressors and use a classifier to partition the data, allowing better modeling of heterogeneous O3 patterns. However, Decision Trees are more interpretable — one can inspect the splits directly.
- **Trade-off:** With vectorised inference, the speed advantage of Decision Trees largely disappears for small test sets. The mixture model with  $n\_components = 4$  dominates on both speed and accuracy in this setting.

#### Conclusion

With vectorised inference, the mixture model with  $n\_components = 4$  achieves the best MAE (10.70 ppb) while remaining fast (1.57 ms). Decision Trees are preferable when model interpretability is critical or when very large test sets make even small per-sample overheads significant. This experiment highlights that the speed disadvantage of mixture models reported in prior (unoptimised) implementations largely disappears with proper vectorisation.

## References

- [1] Kevin P. Murphy. *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012. Chapter 11: Mixture Models and the EM Algorithm.